Doc. 316 – FFGFT and the Riemann Zeta Function

Reach and Limit: Two Identities, Four Exclusions

Johann Pascher

August 2026

Contents

Preliminary: the answer in short . . . 2

.1 Starting point and claim . . . 2

.2 Two findings that hold as identities . . . 2

The Riemann zeta is a factor of the torus spectral zeta . . . 2

The Casimir point lies symmetric to the critical line . . . 3

.3 The decisive obstruction: Weyl against Riemann-von Mangoldt . . . 3

.4 The constructive link – and why it does not carry . . . 4

The approach . . . 4

Why the chain collapses . . . 4

What remains . . . 5

.5 The missing boundary condition – test negative . . . 5

Test discipline . . . 5

Result . . . 5

Why this was structurally to be expected . . . 6

.6 The harmonic approach: ξ as a Tonnetz point . . . 6

The starting observation . . . 6

The responsible point of contact: Weil's formula . . . 6

Measurement . . . 7

The structural reason . . . 7

Does the 5-limit include the higher ones? . . . 8

Measurement instead of assertion: the bicoherence test . . . 9

Can a weighting exist? Three tests . . . 9

.7 Balance . . . 11

.8 Averaging, tempering, resolution: where does the instrument sit? . . . 12

Three domains, three locations . . . 12

The testable form . . . 12

Where the fractal correction stands . . . 12

The actual candidate: cutting off rather than averaging . . . 13

.9 Sources and methodological placement . . . 13

Load-bearing sources . . . 13

Further usable procedures . . . 14

Where the present tests fall short . . . 14

.10 Closing reflection: why one cannot compute backwards . . . 15

What the calculations show . . . 15

The direction that works – and the one that does not . . . 15

Interpretation . . . 16

.11 Open items . . . 16

Note: quantum computers . . . 16

.12 Verification . . . 17

Preliminary: the answer in short

Result

The Riemann hypothesis is an **arithmetic** matter. The torus and ξ contribute nothing to its solution. The result of this document is a *negative* one – three obvious geometric approaches are excluded with reasons given.

The blanket formulation would be too coarse in one respect, hence the refinement.

What remains purely a matter of numbers. Where the zeros lie, how they stand relative to one another, how one grasps them individually – that is carried by arithmetic, by the primes. Three geometric approaches were tested and all are negative: no compact geometry can carry the zeros as its spectrum (Section .3, at theorem level), ξ does not shift the critical line (Section .2), and the ξ ladder discretises nothing (Section .5, out-of-sample test negative).

What is nevertheless not merely a matter of numbers. Two things. First, the Riemann zeta is a *factor* in the spectrum of the $\mathbb{Z}^4$ torus – an identity, not a picture (Section .2). The Riemann hypothesis is thereby a statement about a geometric object of the framework; only one that the geometry cannot answer. Second, the form of the zero density says *where* the question belongs: $N(T)$ grows like $T \ln T$, and no Laplacian on any compact manifold can deliver that – it requires a dilation structure, hence the unrolled domain. The geometry indicates the address without supplying the answer.

The striking thing is precisely that both occur together: the numbers demonstrably sit in the torus spectrum, but the torus cannot generate them.

The yield. Three dead ends are closed, with reasons rather than conjecture. In addition, one of the author's own faulty constructions is documented: a cell argument that initially looked load-bearing turned out on elaboration to be a restatement of $\hbar = 1$ (Section .4) and was withdrawn.

.1 Starting point and claim

[Q] The corpus mentions the Riemann hypothesis in exactly one place: Doc. 024 notes, in an AI/network context, that the "number space" has dimensions "modulated by the Riemann hypothesis and ζ functions". That is a conceptual marginal remark without an epistemic marker and without derivation. A worked-out connection does not exist in the corpus.

[K] At the same time Doc. 314 contains material that touches the ζ function directly: Epstein zeta, Casimir point, heat kernel on T^4. This document examines systematically what follows.

Claim. It proves nothing about the Riemann hypothesis and does not claim to. Three things are clarified: which connections hold as identities **[K]**, which path carries constructively **[B]**, and which obvious paths are *demonstrably* not passable **[X]**. The exclusion is the most valuable part – it saves work on dead ends.

.2 Two findings that hold as identities

The Riemann zeta is a factor of the torus spectral zeta

[K] Doc. 314 carries the closed form of the $\mathbb{Z}^4$ spectral zeta (verified threefold there; confirmed here against the direct sum with analytic tail correction to 10^{-7} through 10^{-12}):

$$Z_{\mathbb{Z}^4}(s) \;=\; 8\left(1 - 4^{1-s}\right)\zeta(s)\,\zeta(s-1).$$

This is not an analogy but an identity. The zeros of the torus spectral zeta therefore have three sources:

Source	Location of the zeros
$\zeta(s) = 0$	$\Re(s) = 1/2$ (non-trivial – the RH)
$\zeta(s-1) = 0$	$\Re(s) = 3/2$ (shifted by 1)
$1 - 4^{1-s} = 0$	$s = 1 + 2\pi i k/\ln 4$ (lattice factor)

On this reading the Riemann hypothesis is a statement about the spectrum of the $\mathbb{Z}^4$ torus – about a geometric object of the framework, not merely about a number series.

The Casimir point lies symmetric to the critical line

[K] Doc. 314 computes the ζ-regularised vacuum energy $E = \pi\zeta_M(-1/2)$ and finds the order reversal: D4 minimises the Epstein zeta for $s > 0$, at $s = 0$ all lattices are equal, and at $s = -1/2$ the $\mathbb{Z}^4$ torus lies lower (-0.9321 against -0.8686).

The Casimir point $s = -1/2$ has, under the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, the mirror point $s = 3/2$. The axis of symmetry:

$$\frac{-1/2 + 3/2}{2} = \frac{1}{2} \qquad \text{– exactly the critical line.}$$

The physically occupied interval $[-1/2,\ 3/2]$ has the critical line as its midpoint: consistency between physics and arithmetic.

[X] No proof follows from this – the critical line is fixed by the functional equation anyway. Nor can ξ shift it: a perturbation of order $\xi \sim 10^{-4}$ (even with the RG amplifier 100ξ) does not reach an axis fixed by a functional identity.

.3 The decisive obstruction: Weyl against Riemann-von Mangoldt

The obvious move would be to seek an operator whose eigenvalues are the zeros (Hilbert-Pólya programme), and to deform $\mathrm{T}^4/\mathbb{Z}_3$ suitably for that purpose. **This entire branch is excluded – at theorem level, not as intuition.**

[Q] *Weyl's law.* For *every* compact Riemannian manifold of any dimension d, asymptotically $N(\lambda) \sim C_d \,\mathrm{Vol}\, \lambda^{d/2}$ – a pure power. Curvature, boundary and orbifold points change only lower orders, not the leading term.

[Q] *Riemann-von Mangoldt (proved).*

$$N(T) = \frac{T}{2\pi} \ln \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O(1/T).$$

[K] The comparison via the local exponent $\mathrm{d}\ln N/\mathrm{d}\ln T$:

T	$N(T)$	local exponent $d/2$
10^3	$6.48 \cdot 10^2$	1.2457
10^5	$1.38 \cdot 10^5$	1.1153
10^8	$2.48 \cdot 10^8$	1.0642
10^{12}	$3.95 \cdot 10^{12}$	1.0403
10^{20}	—	1.0233

The exponent is $1 + 1/\ln(T/2\pi)$ – it falls monotonically towards 1 but is *never constant*. Range over 17 decades: 0.22. The best power fit in the window $[10^3, 10^6]$ deviates systematically by up to $11\,\%$.

Exclusion at theorem level

[X] There is no compact Riemannian manifold – of whatever dimension, curvature, topology or orbifold structure – whose Laplace spectrum is the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

Consequence. The branch "deform $\mathrm{T}^4/\mathbb{Z}_3$ suitably" is settled completely, not only in the flat case. The extra $\ln T$ is the signature of a **scale degree of freedom**: a dilation direction with logarithmically growing volume. That is not a Laplacian on a manifold.

Withdrawn intermediate diagnosis. In a first pass it was argued that $\log(p)$ geodesics require negative curvature. That is wrong: what generates log lengths is a dilation structure (multiplicative group $\mathbb{R}_+^*$), not curvature; the Berry-Keating operator $H = xp$ is the generator of dilation, not of a hyperbolic flow. The Weyl obstruction is the correct and considerably stronger statement, and it renders the curvature question moot.

.4 The constructive link – and why it does not carry

If the zero density requires a scale degree of freedom, the responsible domain is the **unrolled** one (Doc. 315) – multiplicative, logarithmic, with dilation as its natural operation.

The approach

[Q] *Berry-Keating (1999)* [1]. The semiclassical count of the dilation generator $H = (xp+px)/2$ in a phase space with a minimal cell:

$$N(E) = \frac{1}{2\pi}\big[\, E \ln\big(E/(l_x l_p)\big) - E + l_x l_p \,\big].$$

With $l_x l_p = 2\pi$ this becomes exactly the Riemann-von Mangoldt formula (verified: difference constant $0.125 = 1 - 7/8$, a pure boundary term, over $T = 15 \dots 5000$).

The obvious chain would then be: Doc. 314 Ch. D1 carries $\lambda_4 \cdot m = 2\pi$ as an exact position relation; in the unrolled domain the quantity conjugate to scale is the dilation momentum; hence FFGFT supplies the cell condition.

Why the chain collapses

[X] Stage 1 – the cell condition is empty of content. $l_x l_p = 2\pi\hbar$ is, in natural units, the Planck cell, and follows from the canonical quantisation of *every* conjugate pair. That $\lambda_4 m = 2\pi$ has the same form is no independent finding – the de Broglie relation *is* the statement $\hbar = 1$. Moreover the count is logarithmically insensitive to the cell (a factor 2 wrong ⇒ N changes by $\sim 1\,\%$); a "hit" would not have been a test in any case.

[X] Stage 2 – the actual content fails quantitatively. Berry-Keating requires something sharper than the cell: two *independent* cutoffs $x > l_x$ and $p > l_p$ whose product is the Planck cell. The FFGFT double cutoff makes this testable:

Quantity	Value
UV: $L_0 = \xi \ell_P$	$2.155 \cdot 10^{-39}$ m
IR: $L^* = 4\bar{\lambda}_e/\xi^{10}$	$8.698 \cdot 10^{26}$ m
Ratio L^*/L_0	$4.04 \cdot 10^{65}$
required	$2\pi \approx 6.28$
Shortfall	**65 decades**

[X] Stage 3 – a finite window against infinitely many zeros. $\ln(L^*/L_0) = 151.06$, that is 16.93 ξ ladder rungs; the Berry-Keating count within the window gives ~ 1816 modes. A finite scale window carries finitely many modes; the zeros are countably infinite. Even if stages 1 and 2 held, at most an initial segment would result – and an initial segment is not a spectral identity.

Marginal observation, explicitly not booked: 16.93 lies close to 17. Both cutoffs carry their own uncertainties – the prefactor 4 is open per R73, ℓ_P depends on G. Without an error budget this is not assessable under P35.

What remains

To be withdrawn is the formulation "the zero density follows from the FFGFT cell condition". It suggests a finding where a triviality stands.

[B] A structural statement remains, and it is not little: that the counting function has the form $(T/2\pi)\ln(T/2\pi)$ at all requires a **dilation generator**. A Laplacian on a compact manifold cannot deliver it (Section .3). The substantive statement is the *identification of the structure* – the unrolled domain carries the dilation – not the numerical value 2π.

.5 The missing boundary condition – test negative

$H = xp$ has *continuous* spectrum. The counting function fixes the mean density, not the individual zeros. What is missing is the boundary condition that turns the continuum into a discrete spectrum.

The natural candidate. The ξ ladder generates a discrete subgroup $\Gamma_\xi = \{\xi^n\} < \mathbb{R}_+^*$; the quotient $\mathbb{R}_+^*/\Gamma_\xi$ is a *circle* of circumference $L = \ln(1/\xi) = 8.9227$. Admissible dilation frequencies would then be multiples of $2\pi/L = 0.7042$.

Test discipline

[K] Proximity arguments require three controls (P35): a null distribution over random periods in the *same* range; artefact control, because periods larger than the data range automatically produce small scores; and an out-of-sample check with look-elsewhere correction.

The artefact control is not theoretical: a first run reported an apparent signal at period $2\pi/|\ln(74/75)| = 468$. Since all zeros lie < 102, all quotients round to 0 – a pure artefact of a null distribution that did not cover the range. Documented because exactly this error type recurs with ξ proximity arguments.

Result

[K] Candidates within the admissible range (null distribution: mean 0.2502):

Candidate	Period	Score	p value
$2\pi/\ln(1/\xi)$	0.7042	0.2812	0.92
$2\pi/\ln 75$	1.4553	0.2043	**0.005**
$\ln(1/\xi)$	8.9227	0.2891	0.94
2π	6.2832	0.2366	0.18

The borderline candidate $2\pi/\ln 75$ was checked out of sample (zeros 31–50): score 0.2686, $p = 0.81$; look-elsewhere corrected $p = 1.0$.

Boundary condition

[X] The ξ ladder does not supply the missing boundary condition. No ξ-derived period candidate survives the out-of-sample check.

Why this was structurally to be expected

[K] A single-period compactification produces an *equidistant* lattice. The zeros, however, show GUE statistics with level repulsion (Montgomery-Odlyzko [11, 10]): mean normalised spacing 0.9815 (should be ~ 1), fraction of spacings < 0.3 equal to 0.000 – against ~ 0.04 for GUE and ~ 0.26 for a lattice-like spectrum. A circle cannot deliver GUE in principle.

The missing boundary condition is carried by arithmetic (the primes), not by geometry.

.6 The harmonic approach: ξ as a Tonnetz point

The preceding sections treat ξ as a *magnitude* – a parameter whose smallness decides its effect. A second approach asks instead about its *harmonic structure*, that is, about the position of ξ in the prime lattice. It leads to a different point of contact with the theory and yields a sharper exclusion.

The starting observation

[K] The prime factorisation

$$\frac{1}{\xi} = 7500 = 2^2 \cdot 3 \cdot 5^4$$

places ξ as a point in Euler's Tonnetz with coordinates $(a_2, a_3, a_5) = (-2, -1, -4)$. This is the same lattice structure as in the control case of the musical spiral: the primes 2, 3, 5 as axes of the 5-limit space. Equivalently in logarithmic form:

$$\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5 = 8.922658\ldots$$

The harmonic reading suggests a different point of contact than the Laplace operator. A Tonnetz carries three incommensurable logarithmic frequencies $\ln 2$, $\ln 3$, $\ln 5$; three independent frequencies generate a quasi-periodic, precisely not equidistant spectrum. The approach therefore touches the arithmetic of the ζ function directly.

The responsible point of contact: Weil's formula

[Q] Weil's explicit formula couples zeros and primes:

$$\sum_{\rho} h(\gamma) = (\text{archimedean term}) - 2\sum_{p,k} \frac{\ln p}{p^{k/2}}\,\hat{h}(k\ln p).$$

The dual *length spectrum* of the zeros is therefore

$$\mathscr{L} = \{k\ln p \;:\; p \text{ prime},\ k = 1, 2, 3, \ldots\},$$

that is, multiples of logarithms of **single** primes. Whether ξ appears there is directly testable: the Fourier transform of the zero density must show peaks at the lengths in $\mathscr{L}$.

Measurement

[K] Tested is $F(u) = |\sum_n e^{i\gamma_n u}|/N$ for the first 100 zeros; the background is 3000 random u in the same range (median 0.0304, 95 % threshold 0.2005, 99 % threshold 0.2572).

u	$F(u)$	Assignment
1.09861	0.2321	$\ln 3$ (peak, 95 %)
1.60944	0.2651	$\ln 5$ (peak, 99 %)
1.94591	0.2625	$\ln 7$ (peak, 99 %)
2.39790	0.2610	$\ln 11$ (peak, 99 %)
2.56495	0.2676	$\ln 13$ (peak, 99 %)

The primes are visible; the method works. The same measurement for ξ-derived lengths:

u	$F(u)$	Quantity
8.92266	0.1775	$\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5$
4.46133	0.0777	$\ln(1/\xi)/2$
4.31749	0.0384	$\ln 75 = \ln 3 + 2\ln 5$
2.23066	0.0189	$\ln(7500)/4$
0.70418	0.1313	$2\pi/\ln(1/\xi)$

None reaches the 95 % threshold.

Resolution limit. The measurement becomes assessable only once u spreads the phases over the full circle, that is $u \gg 2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.02824$. For $u \to 0$ all phases coincide and $F(u) \to 1$, independently of any structure. Candidates below this limit – for instance $\ln(75/74) = 0.01342$ – are *not measurable* with 100 zeros; they yield neither confirmation nor refutation.

The structural reason

Combination against factorisation

Tonnetz and Euler product share the prime axes but use them in **opposite** ways. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ logarithmises to a sum over prime powers: each prime stands on its own, without cross terms – hence $\mathscr{L}$ contains only $k \ln p$. The Tonnetz lives on the opposite: an interval is $2^a 3^b 5^c$ with several non-zero exponents, and it is precisely the mixing that generates the commas. ξ is such a mixed point.

[X] $\ln(1/\xi)$ is therefore a *sum* of lengths and not a length; sums do not appear as peaks in the trace formula. The nearest genuine length is $3\ln 19 = 8.83332$ at distance 0.08934 – without significance, since proximity within the dense length spectrum says nothing.

Sharpness of the exclusion. It is structural, not a matter of magnitude: it holds unchanged if ξ were of order 1. What matters is not the value of ξ but that $1/\xi$ is composite. This section therefore carries a stronger reason than any estimate based on the smallness of ξ.

Remaining test surface. The *individual* axes $\ln 2$, $\ln 3$, $\ln 5$ are lengths in the Weil spectrum – but they are so for any theory containing primes at all; the factorisation of 7500 selects nothing. A statement would arise only if the *exponents* $(2, 1, 4)$ produced a weighting within the spectrum. The following subsection tests this.

Does the 5-limit include the higher ones?

The limit extension of Test B raises the converse question: can it be shown that the 5-limit already contains the higher limits? The answer comes in five gradations, and the gradation repeats the pattern of the whole document (`z9_limit_einschluss.py`).

[K] (1) Exactly: no, provably. $\ln 7 = a\ln 2 + b\ln 3 + c\ln 5$ with rational a, b, c would mean, after clearing denominators, $7^q = 2^A 3^B 5^C$ – contradicting unique factorisation. This holds for every prime above the limit, hence for infinitely many.

[K] (2) As a dense approximation: yes, arbitrarily well. Since $\ln 2, \ln 3, \ln 5$ are linearly independent over $\mathbb{Q}$, the lattice lies dense in $\mathbb{R}$:

N	Error	Cents	Exponents	Tenney
2	$2.82 \cdot 10^{-2}$	48.77	$(2, 2, -1)$	$5.6 \cdot 10^{-3}$
4	$7.97 \cdot 10^{-3}$	13.80	$(-1, -2, 3)$	$4.4 \cdot 10^{-4}$
6	$4.45 \cdot 10^{-3}$	7.71	$(-5, 2, 2)$	$1.4 \cdot 10^{-4}$
8	$2.29 \cdot 10^{-4}$	0.40	$(1, 7, -4)$	$3.7 \cdot 10^{-7}$

The error of the best approximation $(1, 7, -4)$ is exactly $\ln(4375/4374)$ – **the ragisma**. The coarse approximation $(6, -2, 0) = \ln(64/9)$ has error $\ln(64/63)$, **Archytas' comma**. Both are known from the Euler control case: there as near-closures in the 7-limit, here as the error of the 5-limit approximation to the 7-axis. It is the same quantity read from the opposite direction.

[K] (3) At finite resolution: yes, indistinguishable. The peak width of the measurement in Section .6 is $2\pi/(\gamma_{\max} - \gamma_{\min}) = 0.0021$; the ragisma, at $2.29 \cdot 10^{-4}$, is only 0.11 peak widths:

Location	u	Power
$\ln 7$ exact	1.945910	2.144
approximation $(1, 7, -4)$	1.945682	2.144
coarse $(6, -2, 0) = \ln(64/9)$	1.961659	1.417
control $\ln 7 + 0.05$	1.995910	0.042

The fine approximation is indistinguishable from $\ln 7$, the coarse one falls off, and the control point lies in the noise. At this resolution the 5-limit does effectively contain the 7-axis – a statement about the measuring procedure, not about the arithmetic.

[K] (4) Amplitude-weighted: no, and clearly so. The Tenney weight of the approximation $(1, 7, -4)$ is $3.7 \cdot 10^{-7}$ against $1/7$ for $\ln 7$ as an axis of its own – a factor $3.9 \cdot 10^5$. This also explains *why* the approximation can be so accurate: it requires large exponents, and large exponents mean small weight. Accuracy and weight run opposite to one another.

[B] (5) By tempering: yes, but by stipulation. Declaring the ragisma equal to 1 makes $7 = 2 \cdot 3^7/5^4$ exact in the tempered system; likewise, tempering out Archytas' comma makes the approximation $64/9$ exact.

Inclusion does not arise from structure

The 5-limit does not include the higher limits – unless approximation, limited resolution or tempering is allowed to count as inclusion. This is the same pattern as everywhere in this document: inclusion arises through *rationalisation* (tempering) or through *blur* (resolution), never through structure. Likewise Euler's spiral closes only by tempering, and likewise coupling in the Weil spectrum sits only at exact prime powers (Section .6).

For ξ nothing new follows, but a confirmation: that ξ is a pure 5-limit point cannot be cured by any extension – and the approximations that might replace an extension carry no weight.

Measurement instead of assertion: the bicoherence test

The statement that the Euler product has no cross terms was up to this point argued from the *form* of the product. It can be measured – with the procedure responsible for it and with built-in controls.

Setup. The sequence of zeros is taken as a point measure; its transform in the length variable is $F(u) = \sum_n e^{i\gamma_n u}$. Averaged over height blocks, bispectrum and bicoherence [6] follow as in Section .9. The zeros are computed here via the Riemann-Siegel formula: 2469 of them up to $t = 3000$ (Riemann-von Mangoldt expects 2468.6), residual error $7.5 \cdot 10^{-3}$ in γ, corresponding to 0.02 rad of phase error at $u \approx 2.6$.

Test plan. The Weil spectrum contains $k \ln p$. This yields predictions with positive *and* negative controls: if the sum of two lengths is again a length – that is, a prime power – coupling must appear; if it leads to a composite number, none may appear.

Pre-check. The power of $F(u)$ shows the individual lengths $\ln 2 \ldots \ln 13$ all above the 99 % threshold. The background must be cleaned of the 13 genuine lengths in the window – random u hit one of them with about 2 % probability and otherwise form the outlier tail (median 0.055 against a $99\,\%$ threshold of 0.881 after cleaning).

Result (24 height blocks of 102 zeros each; critical value from 2000 phase-randomised realisations per pair):

Pair	Sum	b^2	$b_c(99\%)$	Verdict	expected
$(\ln 2, \ln 2)$	$\ln 4 = 2^2$	**0.8088**	0.2612	coupling	coupling
$(\ln 3, \ln 3)$	$\ln 9 = 3^2$	**0.8462**	0.2214	coupling	coupling
$(\ln 2, 2\ln 2)$	$\ln 8 = 2^3$	**0.7586**	0.2432	coupling	coupling
$(\ln 2, \ln 3)$	$\ln 6$ comp.	0.0253	0.1832	none	none
$(\ln 2, \ln 5)$	$\ln 10$ comp.	0.0252	0.1912	none	none
$(\ln 3, \ln 5)$	$\ln 15$ comp.	0.0175	0.1718	none	none

Cross terms: measured

Six of six predictions met, with a contrast of a factor 30 to 50 between prime powers and composite numbers. Coupling occurs exactly where the sum of two lengths is again a length, and nowhere else. **[K]** The cross-term statement is thereby measured, not merely argued structurally.

Consequence for ξ. $\ln(1/\xi) = 2\ln 2 + \ln 3 + 4\ln 5$ is a sum over *three different* primes and leads to the composite number 7500. By the pattern measured here, no coupling can sit there – which the independent Fourier finding of Section .6 ($F = 0.1775$, below every threshold) confirms. Open item Z-4 is thereby closed, and the exclusion of Section .6 now rests on a measurement rather than on an argument from form.

Can a weighting exist? Three tests

The remaining test surface read: a statement would arise only if the exponents $(2, 1, 4)$ produced a weighting in the Weil spectrum. Three independent tests decide this question negatively (`z7_gewichtung_unmoeglich.py`).

[K] Test A – weighting strength. Higher harmonics have smaller amplitude; the standard measure is Tenney height, weight $\sim 1/(nd)$. An exponent makes the strength free:

$$w = \left(2^{|a|}\,3^{|b|}\,5^{|c|}\right)^{-\alpha},$$

with $\alpha = 0$ unweighted and $\alpha = 1$ Tenney. The weight sum has the closed form $\prod_p\big[1 + 2p^{-\alpha}/(1-p^{-\alpha})\big]$ and converges for every $\alpha > 0$:

α	Weight sum	eff. point count	Character
0.10	4,284	23,843	blindly dense
0.30	245.8	1,767	middle
0.50	56.9	360	middle
1.00	9.0	36	axes only

Both extremes are unsuitable: unweighted, the lattice hits everything and selects nothing; at Tenney weighting only the axes remain. What matters is the middle around $\alpha \approx 0.3$ with several hundred to a thousand carrying points. The selection test across the whole range:

α	M(Weil)	M(random)	Ratio	p
0.15	2.1205	2.0943	1.013	0.27
0.30	0.5344	0.5210	1.026	0.31
0.50	0.1414	0.1493	0.947	0.64
1.00	0.0114	0.0226	0.503	0.93

No α produces selection; the ratio stays at or below one. The weighting strength is not the problem.

[K] Test B – limit extension. The 5-limit is the simple form of the Tonnetz; extended Tonnetze take in further prime axes, 7 being the next. Tested at $\alpha = 0.3$; primes up to the respective limit are axes and hence excluded as trivial.

Limit	Axes	Points	Ratio	p
5	3	2,197	1.049	0.28
7	4	28,561	1.015	0.38
11	5	371,293	1.048	0.11

Weightier than the p values is the structural point: every extension merely shifts the boundary, it closes no gap.

Limit	trivially covered	entirely absent
5	$2, 3, 5$	all $p \geq 7$
7	$2, 3, 5, 7$	all $p \geq 11$
13	$2 \ldots 13$	all $p \geq 17$

A Tonnetz with limit L covers the finitely many primes $p \leq L$ by definition and the infinitely many $p > L$ not at all – there is nothing in between. Weil's formula, however, sums over all primes. The limiting case settles it: a Tonnetz with infinite limit would no longer be a structure; it would contain every prime as an axis, that is, the Euler product itself – without additional content and without any relation to ξ.

In addition, $\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$ has exponent zero on all axes from 7 onwards. An extension enlarges the lattice but does not change the position of ξ within it; ξ remains a pure 5-limit point.

[K] Test C – rigidity. This is the actual reason, and it holds independently of any numerical computation. The weights of the explicit formula are not free parameters but a consequence of the Euler product:

$$\log \zeta(s) = \sum_{p,k} \frac{1}{k} p^{-ks},$$

from which $\ln p/p^{k/2}$ follows by differentiation and shifting to $s = 1/2$. There is no place at which an additional factor could be inserted without altering the Euler product itself. Setting $\prod_p (1 - g\,p^{-s})^{-1}$ with $g \neq 1$ by way of trial produces a different Dirichlet series with a shifted abscissa of convergence and *without* a functional equation.

Why no weighting is possible

The functional equation $\zeta(s) = \chi(s)\zeta(1-s)$ is precisely what defines the critical line. A weighting meant to explain the location of the zeros thereby destroys the object whose zeros it seeks to explain. **[X]**

Test C carries on its own; A and B show in addition that the numerical routes yield nothing either – neither through the choice of weighting nor through extended Tonnetze. Open item Z-1 is therefore not unanswered but **decided in the negative**.

.7 Balance

Statement	Status
ζ is a factor of the torus spectral zeta	**[K]** verified
Casimir interval has critical line as axis	**[K]**
Zero density requires dilation generator	**[B]** structure
Cell condition as an independent finding	**[X]** only $\hbar = 1$
Double cutoff as BK truncation	**[X]** 65 decades
Compact manifold with zero spectrum	**[X]** Weyl
ξ ladder discretises the zeros	**[X]** test negative
$\ln(1/\xi)$ in the Weil length spectrum	**[X]** a sum, not a length
Cross terms in the Euler product	**[K]** measured by bicoherence
5-limit contains higher limits	**[X]** only approximated/tempered
K_{frak} as tempering instrument	**[X]** no comma in the residue
Weighting by the exponents $(2, 1, 4)$	**[X]** Z-1 decided
ξ shifts the critical line	**[X]** functionally fixed
FFGFT solves the Riemann hypothesis	**[X]**

The most honest sentence. Of the three connections examined, two hold as identities **[K]** – but they are consistency, not a statement about the location of the zeros. The third, the constructive one, collapsed on elaboration: what looked like a derivation is $\hbar = 1$. What remains is a structural statement: the zero density requires a dilation generator, and that lives in the unrolled domain.

What a viable approach would have to achieve: a non-trivial boundary condition on the unrolled scale direction that (a) carries infinitely many modes – hence is not a finite cutoff window – and (b) produces GUE statistics rather than equidistance. Both requirements point in the same direction: the structure must be arithmetic, not geometric.

.8 Averaging, tempering, resolution: where does the instrument sit?

Euler's spiral closes only by tempering (Section .2), and the 5-limit contains the higher limits only by approximation, resolution or tempering (Section .6). In both cases closure arises not from the structure but from something added from outside. It is worth determining what that addition is – for in three domains it sits in three different places.

Three domains, three locations

For hearing, the averaging sits in the *receiver*: critical bands, finite frequency resolution, fusion of closely spaced tones. The measuring instrument becomes part of the result. Tempering here is not an arbitrary stipulation but follows from a property of the listener.

In pure mathematics there is no receiver. The ratios exist regardless of whether anyone distinguishes them. If something is to close that does not close of itself, tempering must be inserted as a *stipulation* – a deliberate intervention, documented in the comma that thereby disappears.

In nature it is open whether there is a receiver. The question therefore arises whether the fractal correction plays that role.

The testable form

[B] A tempering always leaves a residue, and that residue is a *comma*: a ratio of smooth numbers, that is, of products of small primes. The corpus knows one residue – the ≈ 7 eV of P-315-2, $45{,}000$ times the measurement floor. The question is thus testable: does this residue have comma structure?

[K] It does not. The best rational approximations to $1 + 1.36 \cdot 10^{-5}$ are $73530/73529$, $882365/882353$ and $1250017/1250000$; none is smooth. The control probe shows the test's discriminating power: all six commas examined – Pythagorean, syntonic, schisma, breedsma, ragisma, landscape – are recognised as 5- or 7-limit smooth.

On test conduct: without a condition on approximation quality, the smoothness test reports the trivial approximation $1/1$ as "5-limit smooth" – at 100 % residual error. The condition is necessary, not cosmetic; it belongs to the same error class as the artefacts in Sections .5 and .6.

Where the fractal correction stands

K_{frak} is not the tempering instrument

If K_{frak} were a tempering, it would have to leave a comma-like residue. The residue present does not have that structure. **[X]** This sharpens P-315-2: the ≈ 7 eV are more likely a pole-mass effect or a genuine correction term than the by-product of a rationalisation.

[B] A further clarification is easily overlooked: in FFGFT the stipulation sits not at K_{frak} but one level below – at $\xi = 4/30000$ itself. That is the rational intervention; $K_{\text{frak}} = 74/75$ is its *consequence*, not the act. In this respect the fractal correction is not a tempering but the bookkeeping of an already closed cycle.

[B] Nevertheless it sits structurally where an instrument would sit: Doc. 314 Ch. D2 carries K_{frak} as a *property of the traversal*, entering via the bridge constants rather than the local operator. That is literally "the measuring instrument becomes part of the result": K_{frak}

lies not in the field but in the traversing. It resembles a measurement effect in *location*, not in *character* – it is exact, not averaging.

The actual candidate: cutting off rather than averaging

[B] If nature has an instrument, it is more likely the double cutoff. It generates a resolution limit, and precisely there the procedure becomes part of the result – without anything being tempered. Section .6 shows it in a concrete case: at a peak width of 0.0021 the 5-limit approximation to ln 7 is no longer distinguishable from the exact value, although the ragisma between them remains *present*.

> **The difference in one sentence**
>
> A tempering *alters* the values and leaves a comma. A resolution limit alters nothing and leaves nothing – it merely makes differences invisible. Both make distinctions disappear; only the first produces residues.

Hence the tripartition. Hearing averages *actively* and produces commas. Mathematics has no receiver; there tempering must be done by stipulation. Nature, if the framework is right, cuts off *passively*: distinctions vanish without rationalisation – and the fractal correction is not involved in this; the scale limit is.

[B] *Status.* The comma test is an exclusion and rules out one particular reading of K_{frak}. It does not show that no averaging mechanism acts in nature at all – only that the fractal correction is not one. Whether the double cutoff is more than an analogy as an instrument remains open; it depends in turn on quantities the corpus carries as open (the prefactor 4 in L^*, R73).

.9 Sources and methodological placement

The procedures used in this document are standard, and so are their pitfalls. This section places the usable literature and states where the tests computed here fall short of the state of the art.

Load-bearing sources

Berry & Keating (1999) [1]. The basis of Section .4. Comparison of the counting functions suggests that the t_n are eigenvalues of an unknown Hermitian operator obtained by quantising a classical system with Hamiltonian $H_{cl} = xp$. The work also states the requirement on such a system: the "Riemann dynamics" would have to be chaotic and possess periodic orbits *whose periods are multiples of logarithms of the primes*. That is precisely the length spectrum of Section .6 – and the condition on which the Tonnetz fails, since it yields sums rather than multiples.

Sierra (2019) [2] **and Sierra & Townsend (2008)** [3]. These show the state of the Hilbert-Pólya programme: spectral realisations via xp with boundary interaction, via Landau levels, and most recently via a massless Dirac fermion in Rindler spacetime with delta potentials on the square-free integers. Notable for the present question: all these constructions encode the arithmetic *explicitly* (square-free numbers, modular forms, boundary functions from the Riemann-Siegel formula). None derives it from a geometry. This supports the finding of Section .3 from the other side: where the zeros are realised spectrally, it happens by building in number theory, not by choosing a space.

Pápics (2012) [4]. The work methodologically closest to Section .6. It examines whether detected frequencies stand in harmonic relation to one another – the same problem in asteroseismology. Central result: combination frequencies occur in random data with high probability, and as data quality rises the distinction between genuine and spurious combinations becomes *harder*, not easier. The usable discriminator is not the existence of individual combinations but the *number* of lowest-order combinations compared against simulations with matching properties. Exactly this approach – comparison against a simulated null distribution rather than evaluation of individual hits – underlies the permutation tests of this document.

Pólóskei et al. (2018) [5]. Treats the procedure actually responsible for the question of harmonic coupling. Bicoherence

$$b^2(f_1, f_2) = \frac{|B(f_1, f_2)|^2}{A\{|X(f_1)X(f_2)|^2\}\, A\{|X(f_1+f_2)|^2\}}$$

measures not amplitudes but *phase coupling*: whether $f_3 = f_1 + f_2$ occurs with a fixed phase relation $\varphi_3 = \varphi_1 + \varphi_2 + \text{const}$. Decisive for the present use is the warning about false positives: for non-stationary or short signals, classical bicoherence yields high values *without* any coupling. The proposed remedy is a Monte Carlo procedure using the measured Fourier amplitudes with *randomised phases*, from which the null distribution $\rho(b)$ and a critical value $b_c(\alpha)$ are obtained. The trade-off is also stated: the number of averages can be increased only at the cost of frequency resolution ($\Delta f = f_s N/M$).

Further usable procedures

The following methods fall below the Fourier resolution limit by assuming a model (a sum of exponentials) instead of transforming blindly. They have not been employed here but would be the next stage:

- **Subspace methods:** MUSIC (Schmidt), ESPRIT (Roy/Kailath), Min-Norm. They decompose the covariance matrix into signal and noise subspaces.
- **Matrix pencil** [7] for damped and undamped sinusoids.
- **Iterative methods:** RELAX, NOMP; and Volterra-series approaches for the self-consistent removal of combination frequencies (Lares-Martiz et al., MNRAS 2020).
- **Multitaper** (Thomson) with an F-test for line components; **multichannel singular spectrum analysis** for extracting dynamical modes without a Fourier basis.

On the magnitude of the gain: in a published comparison of three frequencies at 440/445/450 Hz over 100 ms, ESPRIT attains a frequency RMSE of 0.087 Hz against 2.88 Hz for spectral MUSIC – the difference between model classes thus exceeds the difference from Fourier analysis.

Where the present tests fall short

[B] The scripts z6 and z7 use the bare Fourier sum $F(u) = |\sum_n e^{i\gamma_n u}|/N$ – the simplest tool available. Three limitations follow:

1. *No coupling measurement*. $F(u)$ measures whether structure sits at length u at all, not whether two lengths are *coupled*. The structural claim that the Euler product has no cross terms is argued in this document from the form of the product. The bicoherence test has since been carried out (Section .6) and confirms it; the remaining tests of this document stay with the simpler procedure.
2. *Statistics*. Sections .6 and .6 compute with 100 zeros from the literature; for bispectra that is too few. Section .6 therefore computes 2469 zeros itself.

3. *Not a time signal.* The sequence of zeros is not a measured signal; stationarity is undefined, and the standard tests for non-stationarity do not apply directly.

These limitations do not touch the **[X]** findings of this document: the Weyl obstruction (Section .3) and the rigidity of the Weil weights (Section .6, Test C) are structural arguments and independent of the choice of spectral method. They concern only the numerical tests, whose force is thereby bounded from below – which does not weaken the negative overall finding but merely qualifies its numerical component.

.10 Closing reflection: why one cannot compute backwards

At the end stands an image that draws the individual findings together, and an interpretation that goes beyond them. The two must be kept apart.

What the calculations show

The primes are axes – in the Tonnetz as in the Euler product. The same numbers, the same lattice structure. For simple ratios the inference back to the underlying harmony is unambiguous: $\ln 2$, $\ln 3$, $\ln 5$ appear as clear lines in the spectrum of the zeros, each in its place, each assignable.

With composition this unambiguity is lost, for two reasons, both visible in the tests.

The first is *ambiguity*. A simple ratio has few possible decompositions. A composite one has many – and the further the lattice is spanned, the denser the possibilities lie, until in the end every number fits somehow. It is not the rarity of a combination that creates the difficulty but its multiplicity of readings: where everything is reachable, reaching says nothing.

The second reason weighs heavier. The Euler product treats each prime *on its own*, without cross terms; its spectrum knows only $k \ln p$, multiples of individual prime logarithms. As soon as a ratio mixes several primes it is no longer a single harmony but a superposition – and superpositions do not appear in the spectrum as lines of their own. $\xi = 2^2/(2^4 \cdot 3 \cdot 5^4)$ is such a superposition.

An image from practice captures it exactly: from the sound of a single reed one can infer the fundamental. In a full chord across several registers one hears the mixture – but the fundamental of the mixture does not exist. There are only the fundamentals of the individual reeds. ξ is a chord, not a note.

The direction that works – and the one that does not

This allows a precise statement of what is actually closed here. It is not the connection between harmony and structure; it is its *reversibility*.

Forwards it carries: from simple ratios the composite can be built, step by step, each step traceable. This is how intervals arise from fifths and thirds, and how ξ arises from 2, 3 and 5.

Backwards it does not carry. Composition is not invertible because it destroys information: many paths lead to the same place, and the path taken can no longer be read off the result. This is not a weakness of method that better techniques could remedy, but a property of the matter itself. The tests of this document confirmed it at three different points: at the weighting strength, at the limit extension, and at the rigidity of the Weil weights.

Interpretation

Interpretation, not a finding

It is natural to suppose that complex formations in nature rest on such harmonies – that the composite is built from simple ratios, as a chord from notes. This conjecture is old and has been fruitful in many respects; it also stands behind the construction of ξ.
What this document shows is something else and more modest: even if that is so, the path cannot be walked backwards. Reconstructing the underlying harmonies from the finished formation is not difficult but impossible – composition has erased the assignment.

The first sentence is an interpretation and remains one; nothing in this document supports it and nothing refutes it. The second is computed.

Between the two lies the actual yield: whoever suspects a harmonic order behind complex structures must establish it *forwards* – by construction and prediction. The way back, inferring the order from the finished structure, is not open. This holds for the Riemann hypothesis, and it holds more generally.

.11 Open items

- **Z-1 (decided, negative):** A weighting of the Weil spectrum by the Tonnetz exponents $(2, 1, 4)$ cannot exist – see Section .6. Neither a weighting strength nor a limit extension produces selection; above all, the weights of the explicit formula are not free parameters.
- **Z-2:** The lattice factor $(1 - 4^{1-s})$ yields zeros at $s = 1 + 2\pi ik/\ln 4$. Does this ladder have a corpus meaning (D4? the factor 4 from $L^* = 4\bar{\lambda}_e/\xi^{10}$, R73)? Not yet examined.
- **Z-3:** The heat-kernel finding of Doc. 314 (D4/$\mathbb{Z}^4$ difference exponentially small, $1.8 \cdot 10^{-13}$ at $t = 0.1$) has a counterpart in ζ theory (theta-series fine structure). Whether a test surface follows: open.
- **Z-4 (closed):** the bicoherence test has been carried out (Section .6) and confirms the cross-term statement in six of six controls.
- **Not recommended:** further search for ξ periods in the zeros. Section .5 excludes this structurally via the GUE argument, not merely numerically.

Note: would quantum computers help here?

[B] The question is natural, all the more since Doc. 024 mentions the ζ function in connection with factorisation and Shor's algorithm. The answer is negative, and the reason is the same as in the preceding sections.

Quantum computers accelerate *search problems*: Shor factorises quickly, Grover searches faster. Both presuppose that there is something to find – a factor, a marked entry – and that computing time is the bottleneck. Precisely this is not the case here:

- The Weil weights are not free parameters. There is no search space in which to search; searching faster for something non-existent leads nowhere.
- Composition destroys information. That is not a complexity problem but a loss of information – a quantum computer likewise cannot reconstruct what is no longer there.
- A Tonnetz of finite limit omits infinitely many primes. More computing power pushes the limit further and still omits infinitely many.

There is also the limit of all numerics in this question: zeros can be computed further up than the currently known orders of magnitude, but each further one is an additional data point for a statement ranging over *all* zeros. Numerics can refute the Riemann hypothesis – one counterexample suffices – but can never prove it.

The mention in Doc. 024 links two things that are associatively but not substantively connected: factorising a *particular number*, where quantum computers genuinely achieve something, and a statement about the structure of *all* primes, where they do not.

.12 Verification

All numbers are reproducible by ten scripts (standard library only, target values as assertions, seed 20780458):

- `z1_weyl_obstruktion.py` – Weyl's law against Riemann-von Mangoldt, local exponent, power fit.
- `z2_zellbedingung.py` – Berry-Keating count, equivalence to the Riemann form, cross-check against Weyl.
- `z3_randbedingung_test.py` – ξ ladder test with artefact control, null distribution, out-of-sample, GUE control.
- `z4_epstein_casimir.py` – factorisation against the direct sum, Casimir symmetry point.
- `z5_zellbedingung_kritik.py` – collapse of the cell argument in three stages.
- `z6_tonnetz_weil.py` – Tonnetz coordinates of ξ, Fourier detection of the Weil length spectrum, resolution limit.
- `z10_kommatest.py` – does the residue of P-315-2 have comma structure? Smoothness test with control probe.
- `z9_limit_einschluss.py` – five readings of limit inclusion; ragisma and Archytas' comma as approximation errors.
- `z8_bikohaerenz.py` – zeros via Riemann-Siegel, bicoherence with phase randomisation, positive and negative controls.
- `z7_gewichtung_unmoeglich.py` – weighting strength α, limit extension (5-, 7-, 11-limit), permutation tests, rigidity of the Weil weights.

Inputs: $\xi = 4/30000$ exact; $\ell_P = 1.616255 \cdot 10^{-35}$ m, $\bar{\lambda}_e = 3.8615926796 \cdot 10^{-13}$ m (CODATA 2022); zero values (first 50) from the literature.

Bibliography

[1] Berry, M. V., & Keating, J. P. (1999). *The Riemann Zeros and Eigenvalue Asymptotics*. SIAM Review, 41(2), 236–266. https://doi.org/10.1137/S0036144598347497

[2] Sierra, G. (2019). *The Riemann Zeros as Spectrum and the Riemann Hypothesis*. Symmetry, 11(4), 494. https://doi.org/10.3390/sym11040494

[3] Sierra, G., & Townsend, P. K. (2008). *Landau Levels and Riemann Zeros*. Physical Review Letters, 101, 110201. https://doi.org/10.1103/PhysRevLett.101.110201

[4] Pápics, P. I. (2012). *The puzzle of combination frequencies found in heat-driven pulsators*. Astronomische Nachrichten, 333(10), 1053–1056. https://doi.org/10.1002/asna.201211815

[5] Pólóskei, P. Z., Papp, G., Por, G., Horvath, L., & Pokol, G. I. (2018). *Bicoherence analysis of nonstationary and nonlinear processes*. arXiv:1811.02973 [eess.SP]. https://arxiv.org/abs/1811.02973

[6] Kim, Y. C., & Powers, E. J. (1979). *Digital bispectral analysis and its applications to nonlinear wave interactions*. IEEE Transactions on Plasma Science, 7(2), 120–131.

[7] Hua, Y., & Sarkar, T. K. (1990). *Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise*. IEEE Transactions on Acoustics, Speech and Signal Processing, 38(5), 814–824.

[8] Lares-Martiz, M., Garrido, R., & Suárez, J. C. (2020). *Self-consistent method to extract non-linearities from pulsating star light curves – I. Combination frequencies*. Monthly Notices of the Royal Astronomical Society, 498(4), 6069–6086. https://doi.org/10.1093/mnras/staa2762

[9] Euler, L. (1739). *Tentamen novae theoriae musicae ex certissimis harmoniae principiis dilucide expositae*. Petropoli: Ex Typographia Academiae Scientiarum.

[10] Odlyzko, A. M. (1987). *On the distribution of spacings between zeros of the zeta function*. Mathematics of Computation, 48(177), 273–308.

[11] Montgomery, H. L. (1973). *The pair correlation of zeros of the zeta function*. Analytic Number Theory, Proc. Sympos. Pure Math. XXIV, 181–193. American Mathematical Society.

[12] Tiesinga, E., Mohr, P. J., Newell, D. B., & Taylor, B. N. (2024). *CODATA Recommended Values of the Fundamental Physical Constants: 2022*. Reviews of Modern Physics, 97, 025002. https://physics.nist.gov/cuu/Constants/

[13] Pascher, J. (2026). *FFGFT – Fundamental Fractal Geometric Field Theory*. Corpus, in particular Docs. 230/231/232 (Hilbert space), Docs. 295/313 (closure fork), Doc. 314 (D4 lattice, Epstein zeta, Casimir), Doc. 315 (rolled up/unrolled), Doc. 190 (correction register). https://github.com/jpascher/T0-Time-Mass-Duality